

Alignment Is Not Dependence: Measuring and Reducing Unimodal Shortcuts in Higher-Order Multimodal Fusion

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Abstract

Higher-order multimodal methods fuse representations of several modalities and align them contrastively, on the premise that fused representations capture information unavailable to any constituent modality. We show this premise is empirically unfounded. Across three established benchmarks (AV-MNIST, CMU-MOSI, UR-FUNNY), interaction representations that are active, predictive, high-rank, and permutation-sensitive are almost entirely predictable from a single modality (nonlinear predictability up to $R^2 = 0.989$). We formalize the resulting *alignment-dependence gap* and introduce an operational diagnostic suite that separates fusion, dependence, and conditional utility. We then propose **S1**, a training-time adversarial de-shortcutting objective that penalizes predictability of each interaction from each single modality while leaving joint predictability unconstrained. S1 is the only objective among those studied that significantly reduces the measured shortcut (paired $p \approx 0.011$, five seeds) at zero accuracy cost, and it fabricates no dependence on negative controls. On a new controlled synthetic benchmark with ground-truth third-order synergy, order-restricted models sit at chance while an order-matched interaction solves the task with perfectly balanced modality dependence (+49.9 pp, 5/5 seeds). Finally, on a newly released natural benchmark with genuine headroom (Bird-MML), the original higher-order objective captures none of the available complementarity (+0.2 pp of +15.9 pp available), while our utility+S1 stack recovers a significant share (+5.8 pp over baseline, paired $p < 0.001$). We release the diagnostic suite, the synthetic benchmark, and all experimental logs.

1 Introduction

Multimodal learning increasingly relies on fusing per-modality encodings into joint representations, aligned with contrastive objectives (CLIP-style and higher-

order variants). The implicit claim: the fused representation *integrates* the modalities. We ask a simpler question that the literature rarely answers directly — does the fused representation actually *depend* on all of its inputs?

Our central finding is a repeatable negative: fused interactions can satisfy every surface health criterion (variance, effective rank, standalone probes, permutation sensitivity) while being a unimodal function. We make this precise with predictability probes: on AV-MNIST, $R^2(r_1 \rightarrow r_{12}) = 0.989$ while $R^2(r_2 \rightarrow r_{12}) = 0.031$.

Contributions. (1) We formalize the alignment-dependence gap and release an operational diagnostic suite (§3). (2) We audit three standard benchmarks and show their higher-order interactions are shortcut-dominated (§4). (3) We propose S1, an adversarial de-shortcutting objective with a hinged, joint-sparing penalty; it is the only tested objective that significantly reduces single-modality predictability at no accuracy cost, and it is honest on negative controls (§5). (4) We contribute a controlled synthetic order benchmark on which order-matched interactions are provably (empirically) necessary and sufficient, while order-restricted decompositions sit at chance (§6). (5) We document two methodology pitfalls: post-aggregation LayerNorm leaks higher-order signal, invalidating “low-order” ablations; and online adversaries equilibrate far below converged probes (the *adversary gap*) (§9).

2 Related Work

Contrastive multimodal alignment. CLIP and its tri-modal extensions (Symile, TriCLIP, ImageBind) align embeddings pairwise or via total correlation. ConFu (our direct baseline) aligns fused pair representations with the remaining modality. None of these objectives constrains the *content* of the fused representation.

Modality dominance. Multimodal networks overfit (Wang et al.), gradient rebalancing (OGM-GE and follow-ups), and underspecification (D’Amour et al.) address dominance at the loss/gradient level. We show why loss-level pressure cannot bootstrap dependence: corrupted and positive interactions coincide under a shortcut.

Synergy and information decomposition. PID (Williams–Beer) and MultiViz-style interaction auditing motivate our vocabulary; our metrics are operational approximations, not information-theoretic quantities.

Adversarial representation control. Domain-adversarial training and “learning-not-to-learn” remove unwanted predictability. S1 differs in target: predictability of the *interaction* from *each constituent alone*, never from the pair jointly.

3 The Alignment–Dependence Gap

Setup. Encoders E_i produce $r_i = E_i(X_i)$; a fusion module produces $r_{12} = F(r_1, r_2)$. We distinguish four properties: *active* (non-trivial variance), *predictive* ($I(Y; r_{12}) > 0$), *dependent* (performance degrades when either constituent is corrupted), and *conditionally useful* ($\text{Perf}(r_1, r_2, r_{12}) > \text{Perf}(r_1, r_2)$). Only the combination of the last two earns the name *complementary*.

Diagnostics. (i) zero vs. shuffle interaction tests; (ii) per-modality corruption drops D_1, D_2 and the balance score $B_{12} = 2 \min(D_1, D_2) / (D_1 + D_2)$; (iii) capacity-matched conditional linear and nonlinear probes; (iv) correction–regression counts against the lower-order model; (v) nonlinear predictability probes $q_1(r_1) \rightarrow r_{12}$, $q_2(r_2) \rightarrow r_{12}$, $q_{12}(r_1, r_2) \rightarrow r_{12}$ with the shortcut criterion $R^2(q_1) \gg R^2(q_2)$ and $R^2(q_1) \approx R^2(q_{12})$.

4 Auditing Existing Benchmarks

Every audited benchmark fails either the dependence gate (a modality shuffle is free) or the utility gate (lower-order probes already saturate). These are *data properties*; no training objective can create the missing information, and we log all failed attempts as negative results.

Table 1: Five-seed audits. Interactions are healthy on the surface and shortcut underneath.

Benchmark	Surface health	True content
AV-MNIST	rank $\uparrow$, shuffle-sensitive	$R^2(r_1 \rightarrow r_{12}) = 0.989$, audio drop
CMU-MOSI	text dominance	pair conditional gains $-0.96/$
UR-FUNNY	high rank, sensitive	$R^2 \approx 0.97$; pair gains $-0.66/+0.$

5 S1: Adversarial De-Shortcutting

Mechanism. For each interaction $t = \text{std}(r_{12})$ (EMA statistics, stop-grad), adversaries q_1, q_2 are trained to predict t from r_1 alone and r_2 alone (alternating minimax, $k_{\text{adv}}=1$). The generator pays the hinged penalty $\text{ReLU}(\tau - \text{MSE}(q_i(r_i), t))$, taxing single-modality R^2 above $1-\tau$ without demanding noise. Joint predictability from (r_1, r_2) is never penalized. Per-factor variance floors close the constant-factor escape route.

Results (AV-MNIST, five seeds, paired). Accuracy 70.05 ± 0.48 vs. 69.96 ± 0.59 (ablation, ns); $R^2(r_1 \rightarrow r_{12})$: $0.989 \rightarrow 0.682$ ($t = -4.43$, $p \approx 0.011$); audio dependence doubles ($+0.084 \rightarrow +0.236$ pp, $p \approx 0.059$). S1 is the first objective in this study to move single-modality predictability at all; utility-loss (P1) and ranking (P2) variants never did.

Honesty. On synthetic no-synergy controls and on a distractor-modality control, S1 fabricates nothing (drops ≈ 0 ; gate stays silent). On AV-MNIST the residual $R^2 \approx 0.68$ is an adversary gap (§9), not hidden dependence.

6 The Controlled Order Benchmark

Three (to six) modalities observe latent bits through fixed random prototypes plus Gaussian noise; labels are $y = \bigoplus_i x_i$ (order M) or $y = x_1 \oplus x_2$ with distractors (control). Probes use held-out splits; five seeds per cell.

Table 2: Order-3 synergy ($y=a \oplus b \oplus c$): accuracy (5 seeds).

Model	orders	accuracy
Additive	1	0.496 ± 0.006
Pairs-only	1+2	0.501 ± 0.010
Concat-MLP	joint	1.000 ± 0.000
Full (order-matched)	1+2+3	1.000 ± 0.000

The M -ladder ($M=3..6$) confirms the pattern at

every M (order-restricted: chance; order-matched: 1.0000; concat-MLP cracks at $M=6$, 0.9975). The full model’s dependence is maximally balanced (every single-modality shuffle costs ≈ 0.5 accuracy), and held-out probes of singles/pairs sit at chance. Notably, pair representations do contain the bits (concat probe 0.97) — the order-2 failure is a *readout* failure of sum aggregation, evidence that higher-order synergy requires order-matched interaction *computation*, not merely rich features.

7 Bird-MML: A Natural Benchmark with Genuine Headroom

Bird-MML (149 species; image, audio, text) is audited on frozen pretrained features with a fixed stratified split (71-species subset; the Zenodo release is missing one archive part — all reported factors are identical across compared methods). The audit finds exactly one strong pair: image+text reaches 80.89% vs. best single 64.95% (+15.94 pp headroom), while audio carries nothing in speech-domain features.

Table 3: Bird-MML, five seeds paired. ConFu++ = utility objective + S1.

	ConFu (orig.)	ConFu+U	ConFu+U+S1
probe full acc (%)	76.90 $\pm$ 0.65	83.01 $\pm$ 0.13	82.66 $\pm$ 0.37
conditional gain (pp)	+0.19	+1.07	+0.77
$R^2(\text{text} \rightarrow z_{13})$	0.59	0.44	0.50
audio shuffle drop (pp)	+0.01	+0.01	+0.01

The original objective leaves the available complementarity unused (full $\approx$ lower). ConFu++ recovers +5.76 $\pm$ 0.86 pp over it (paired $t=14.91$, $p<0.001$), exceeding the raw-feature linear ceiling (81.01%), while fabricating no dependence for the null modality.

Fair-play controls. (i) **Leakage control:** 33% of released captions contain the species name; re-auditing with BLIP-only (name-free) captions reduces the headroom from +15.9 to +4.8 pp — still positive, so genuine complementarity exists beyond leakage. (ii) **Attribution:** the accuracy gain comes from the utility objective (ConFu+U 83.01 $\pm$ 0.13 vs. +S1-only 76.58 $\pm$ 0.51); S1’s role is representation quality, not score — on the balanced pair, text-predictability of z_{13} is lowest under the utility objective (R^2 0.59 $\rightarrow$ 0.44) and S1 pins it to the hinge band (0.50 at $\tau=0.5$). (iii) **Same-topic baseline:** OGM-style gradient rebalancing on identical frozen fea-

tures does not beat plain joint training (80.68 $\pm$ 0.38 vs. 80.88 $\pm$ 0.11, paired $t=-1.16$, ns) and sits 1.8 pp below ConFu++. (iv) **Hyperparameter robustness:** S1 is insensitive on ground-truth synergy (synthetic Config A: acc $\approx$ 1.0, balanced drops, for $\tau \in \{0.3, 0.5, 0.7\}$, $k_{adv} \in \{1, 3\}$).

8 End-to-End Replication

To verify the frozen-feature findings transfer, we train encoders, fusion, and objectives jointly on the same subset (identical initialization, schedule, and inference capacity across arms; three seeds). Every arm converges to **text dominance** (text shuffle drop +4.9 pp, image $\approx$ 0, audio 0), because the released captions contain species names (33%) and the task loss absorbs the leakage. Accuracy is statistically flat across arms (confu 71.87 $\pm$ 0.16, +U 72.18 $\pm$ 0.48, +U+S1 72.32 $\pm$ 0.62; paired +0.45 $\pm$ 0.49 pp, ns), while the shortcut meters respond monotonically as designed: $R^2(\text{text} \rightarrow z_{13})$ falls 0.81 $\rightarrow$ 0.73 $\rightarrow$ 0.68 across confu $\rightarrow$ +U $\rightarrow$ +U+S1. End-to-end, the alignment-dependence gap manifests as *modality abandonment* — a stronger failure mode than the frozen-feature shortcut, and exactly what our diagnostics are built to expose.

9 Methodology Pitfalls

Post-aggregation LayerNorm is a hidden higher-order term. A final LayerNorm divides by a cross-modal statistic; additive models ending in LayerNorm reach 0.75 on 3-bit parity (exactly 0.50 without). “Low-order” ablations must use LN-free readouts. **The adversary gap.** An online adversary updated once per batch equilibrates at $R^2 \approx 0.3$ –0.4 (below our hinge band) while a converged offline probe reads 0.84; de-shortcutting claims must therefore be measured with converged probes, not training-time meters.

10 Discussion

The causal chain is now complete: where higher-order information is absent or dominated (AV-MNIST, MOSI, UR-FUNNY), interactions shortcut; where it is present and observable (synthetic order benchmark), order-matched interactions extract it completely and S1 neither breaks nor fabricates anything. The immediate future work is a natural benchmark with validated headroom (Bird-MML), where

the same audit-then-de-shortcut pipeline applies unchanged. **Limitations.** S1’s hinge tolerates redundancy by design (τ); pair-conditional utility of individual interaction terms remains negative (utility arrives via the joint head); the Bird-MML subset uses frozen features (an end-to-end replication trains overnight at writing) and the audio channel is null under speech-domain features (an AudioSet-domain re-extraction is in progress); caption name leakage is documented (33%) with a leakage-free control arm. All predictability statements are operational approximations.

Reproducibility

Five seeds, paired statistics, capacity-matched probes, train-only parameter accounting, git-state logging per run, and public release of the diagnostic suite, synthetic benchmark, configs, and experiment logs.

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